

# E-commerce Sales Data Analysis Using Python

## Internship Final Project Report

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Organization: Upskill Campus & UniConverge Technologies Pvt Ltd (UCT)

Domain: Python Development

Duration: 6 Weeks

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### Executive Summary

This report presents the final project undertaken during my internship with Upskill Campus in collaboration with UniConverge Technologies Pvt Ltd (UCT).

The project, titled “E-commerce Sales Data Analysis Using Python”, focuses on analyzing transactional sales data through Python libraries such as NumPy, Pandas, and Matplotlib. The goal of this project was to provide a structured, automated, and scalable way of cleaning data, generating insights, and creating visualizations that support business decision-making.

Through this project, I was able to:

Perform data cleaning and preprocessing on raw datasets.

Derive business KPIs like total revenue, number of orders, number of customers, and average order value.

Identify top-selling products and analyze monthly revenue trends.

Visualize sales distribution across countries.

Generate structured reports and graphical insights automatically.

This internship and project provided me with hands-on industry-level experience in Python programming, problem-solving, and professional reporting. It strengthened my foundation for future roles in data science, analytics, and software development.

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## Preface

This internship served as a bridge between my academic learning and industry practices. Throughout the six weeks, I focused on strengthening my understanding of Python fundamentals, data analysis, and visualization.

In the final project, I implemented a complete end-to-end data analysis pipeline, starting from dataset ingestion to generating outputs in the form of structured reports and visualizations.

I am grateful to Upskill Campus and UniConverge Technologies Pvt Ltd (UCT) for providing me this opportunity. I also thank my mentors for their support and guidance. This project not only improved my technical expertise but also gave me confidence in applying concepts to real-world problems.

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## Introduction

### About UniConverge Technologies Pvt Ltd

UniConverge Technologies (UCT) is a technology-driven company specializing in IoT, cloud solutions, and custom software development. UCT provides students with industry-level projects and real-world exposure through training programs in collaboration with institutions like Upskill Campus.

### About Upskill Campus

Upskill Campus is a career development and training platform that bridges the gap between academic learning and industry requirements. It provides internships, mentorships, and real-time projects to students, enhancing their practical skills and employability.

## Objective of the Project

The key objectives of this project were:

1. To apply Python programming for real-world data analysis.
2. To implement data cleaning, transformation, and visualization using NumPy and Pandas.

3. To automate business KPI generation for better decision-making.
4. To gain hands-on experience in project development, GitHub usage, and report preparation.

## References

Learning Python by Mark Lutz (O'Reilly)

Python official documentation

Stack Overflow & online coding communities

Tutorials on NumPy, Pandas, and Matplotlib

## Glossary

KPI (Key Performance Indicator): Metrics used to measure business performance.

AOV (Average Order Value): Average revenue per order.

EDA (Exploratory Data Analysis): The process of analyzing datasets to summarize their main characteristics.

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## Problem Statement

Businesses collect huge volumes of sales data daily. Traditional methods like Excel spreadsheets or manual reporting are:

Time-consuming

Error-prone

Not scalable for large datasets

The problem addressed in this project is to design a Python-based automated system that can:

Clean raw sales data

Generate key performance indicators (KPIs)

Identify sales trends and top products

Provide visual insights for decision-making

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Existing and Proposed Solution

Existing Solutions

Manual analysis using Excel or spreadsheets

Limited automation, scalability, and efficiency

Difficulties in handling large datasets

Proposed Solution

The proposed system uses Python libraries to automate and improve the process:

NumPy: For numerical computations and efficient array operations

Pandas: For data cleaning, wrangling, and analysis

Matplotlib: For generating visualizations of KPIs and trends

The Python script automatically:

Reads the dataset (.csv file)

Cleans and preprocesses data

Generates KPIs (Revenue, Orders, Customers, AOV)

Plots graphs (Monthly Revenue, Top Products, Country Sales)

Saves results into reports and charts for further usage

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Proposed Design/Model

High-Level Workflow Diagram

Input Dataset   Data Cleaning (Pandas)   KPI Calculation (NumPy)   Visualization (Matplotlib)  
Reports & Outputs

Low-Level Design Steps

1. Data Ingestion: Load dataset from .csv file.
2. Cleaning & Preprocessing: Handle missing values, remove invalid records, compute total price.
3. Feature Engineering: Extract useful time-based features like month, year, weekday.
4. KPI Analysis: Calculate total revenue, AOV, customers, top products, monthly sales.
5. Visualization: Create bar charts, line charts, and pie charts for easy interpretation.
6. Output: Save KPIs in JSON, summary report in Markdown, and charts as PNG images.

Interfaces

Command Line Interface (CLI):

```
python final_project.py --input data/ecommerce_sales_sample.csv --output results/
```

Outputs stored in the results/ folder:

kpis.json

summary.md

monthly\_revenue.png

top\_products.png

sales\_by\_country.png

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Performance Test

Test Plan

The project was tested using sample datasets to ensure functionality across all modules.

Test Cases

| Test Case | Input | Expected Output | Status |
|-----------|-------|-----------------|--------|
|-----------|-------|-----------------|--------|

|              |                            |                             |      |
|--------------|----------------------------|-----------------------------|------|
| Load Dataset | ecommerce_sales_sample.csv | Dataset loaded successfully | Pass |
|--------------|----------------------------|-----------------------------|------|

|               |             |                                 |      |
|---------------|-------------|---------------------------------|------|
| Data Cleaning | Raw dataset | Clean dataset (no null/invalid) | Pass |
|---------------|-------------|---------------------------------|------|

|                 |                 |                                 |      |
|-----------------|-----------------|---------------------------------|------|
| KPI Calculation | Cleaned dataset | Correct values for Revenue, AOV | Pass |
|-----------------|-----------------|---------------------------------|------|

|                       |                 |                         |      |
|-----------------------|-----------------|-------------------------|------|
| Top Products Analysis | Cleaned dataset | List of top 10 products | Pass |
|-----------------------|-----------------|-------------------------|------|

|               |                 |                                  |      |
|---------------|-----------------|----------------------------------|------|
| Visualization | Cleaned dataset | PNG charts generated in results/ | Pass |
|---------------|-----------------|----------------------------------|------|

Test Procedure

Run the script with a sample dataset.

Validate each output file in the results folder.

## Performance Outcome

The system successfully executed all steps, generating correct KPIs and visualizations.

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## My Learnings

Through this internship and final project, I gained:

Strong understanding of Python fundamentals.

Hands-on experience with NumPy and Pandas for data manipulation.

Ability to clean, preprocess, and analyze real-world datasets.

Practical exposure to Matplotlib for visualization.

Knowledge of GitHub repository management and project documentation.

Improved skills in debugging, problem-solving, and reporting.

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## Future Work Scope

This project can be extended further:

1. Integration with advanced visualization libraries like Seaborn or Plotly.
2. Implementing machine learning models for predictive sales analysis.
3. Scaling the project using big data frameworks like PySpark or Dask.

4. Deploying the project as a web dashboard (using Flask/Django + React).